

CRC Tumour-Region Detection -- Performance Report

3-model ensemble + Macenko stain normalisation | threshold 0.4 | 2026-09-21

section	ground truth	tissue tiles	tumour area	mean P(TUM)	regions exported	result
Cancer_P1	tumour	2,717	32.28%	0.280	2	detected
Cancer_P2	tumour	2,821	48.49%	0.365	1	detected
Cancer_P5	tumour	2,794	27.67%	0.263	5	detected
Normal_P3	normal	2,454	0.08%	0.021	0	clean
Normal_P5	normal	2,365	1.82%	0.040	0	clean

Summary

Tumour sections mean tumour area 36.1% (range 27.7% - 48.5%)
Normal sections mean tumour area 0.9% (range 0.1% - 1.8%)
Separation +25.8% (min tumour - max normal; > 0 = groups do not overlap)
Negative control 2 of 2 healthy sections produced ZERO exported tumour regions

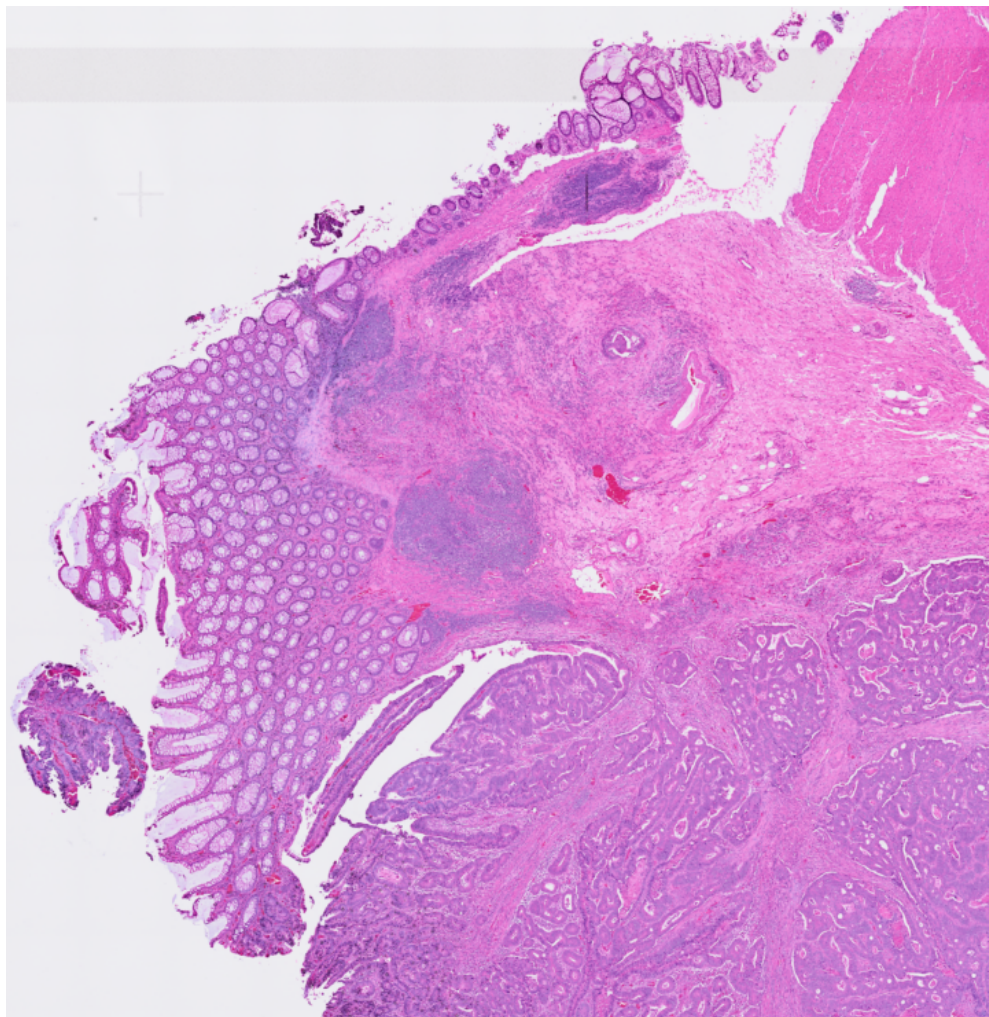
Validation against transcriptomic ground truth (Cancer_P1, independent modality)
2,307 tiles scored against Visium HD cell-type labels (859 molecularly tumour-positive)
AUROC 0.9855 (threshold-free)
At the same threshold 0.4 used for the areas and boundaries above:
precision 0.935 recall 0.921 specificity 0.962
The image model never sees gene expression, so this is an independent check.

How to read this report

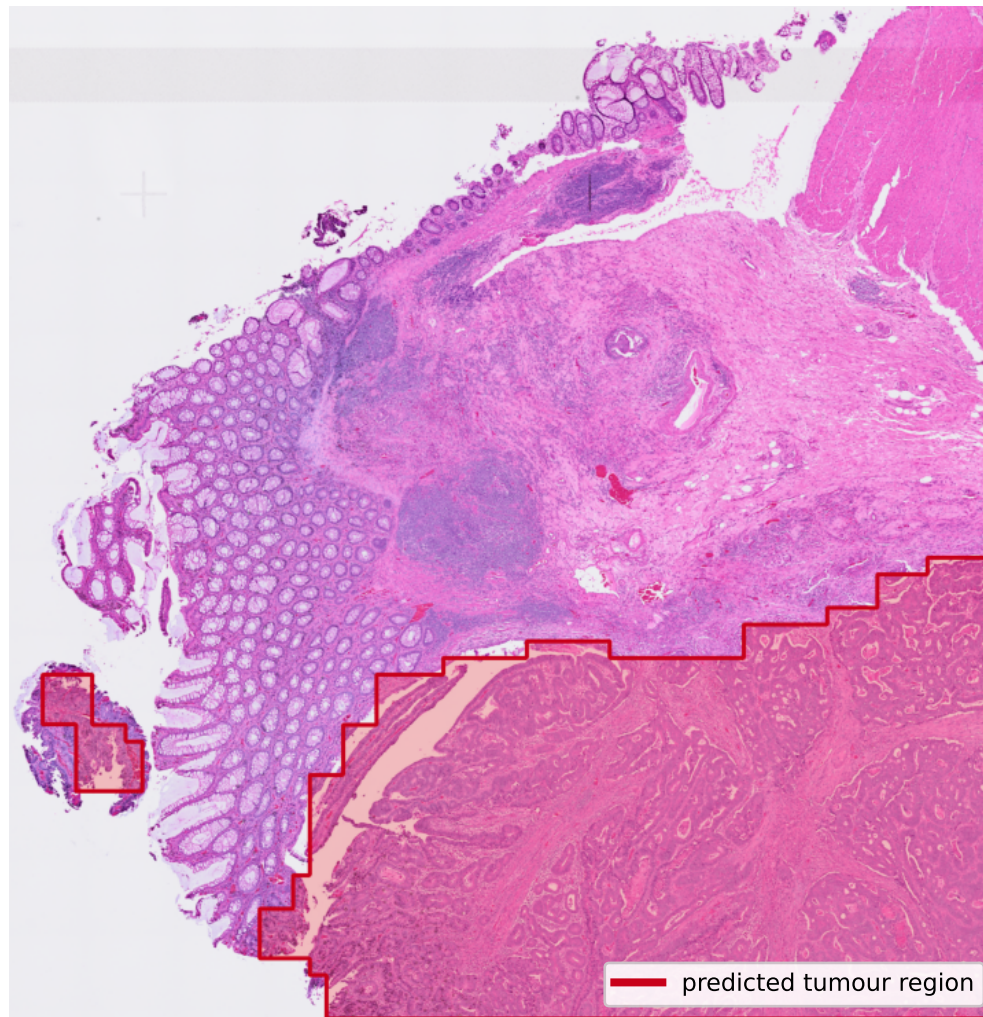
A negative control is necessary but not sufficient: a model predicting 'no tumour' everywhere would also pass it. The evidence rests on both sides holding at once -- tumour found in the cancer sections AND nothing called in the healthy ones. Boundaries are drawn at 224 px (112 um) tile resolution, so the red line is a tumour-REGION estimate, not a histologically precise margin. Regions contain stroma, immune and vascular cells as well as malignant glands. The 0.4 operating point was chosen on this same Cancer_P1 molecular data, so the precision/recall above are optimistic; the AUROC is threshold-free and unaffected. Research pipeline, not a diagnostic device. Not validated for clinical use.

Cancer_P1 | tumour area 32.3% | 2,717 tissue tiles | exported regions 2 | correctly outlines tumour

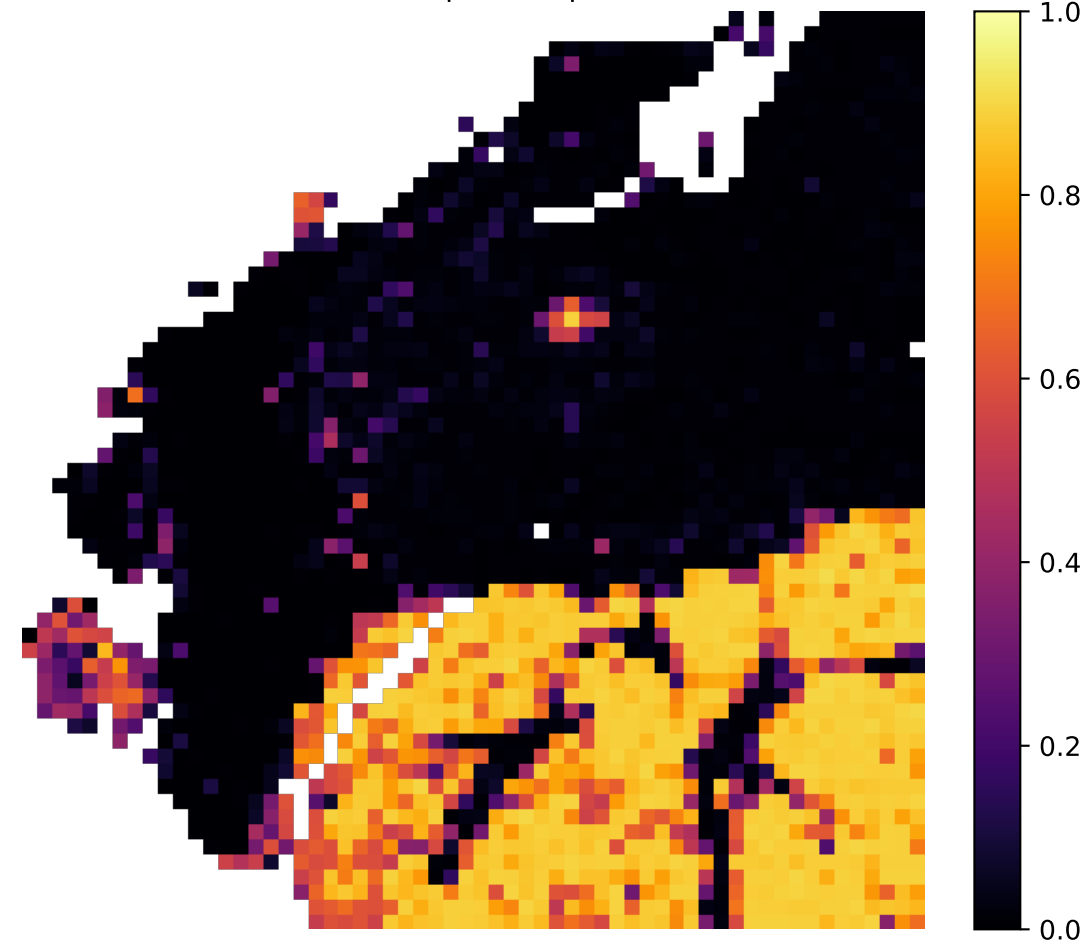
Cancer_P1 -- H&E (ground truth: tumour)



tumour boundary at $P(\text{TUM}) \geq 0.4$ -- 2 region(s)

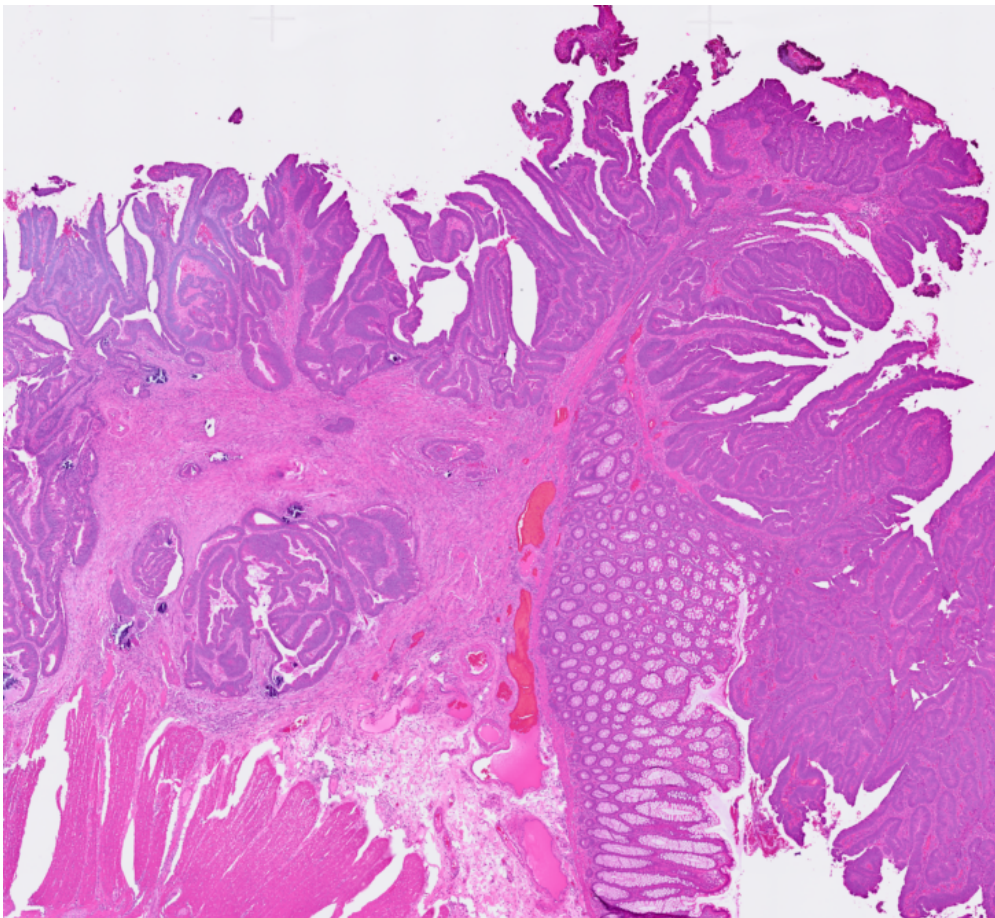


$P(\text{tumour})$ per 224 px tile

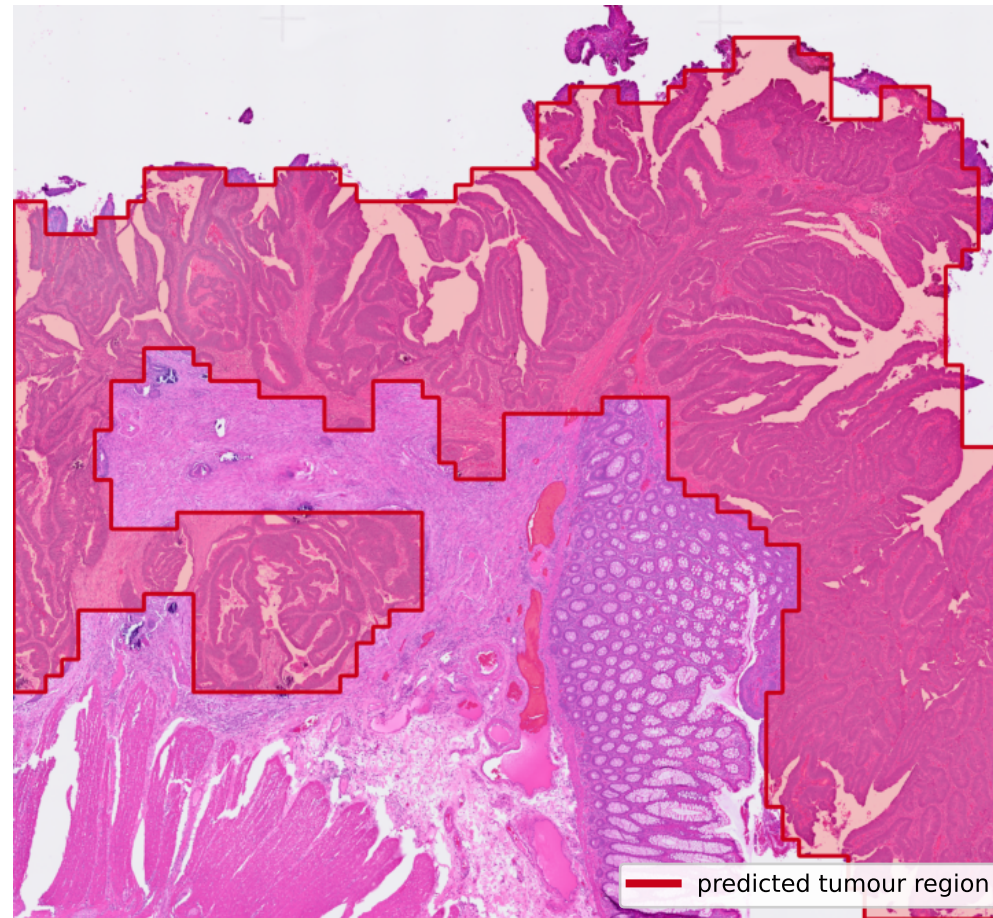


Cancer_P2 | tumour area 48.5% | 2,821 tissue tiles | exported regions 1 | correctly outlines tumour

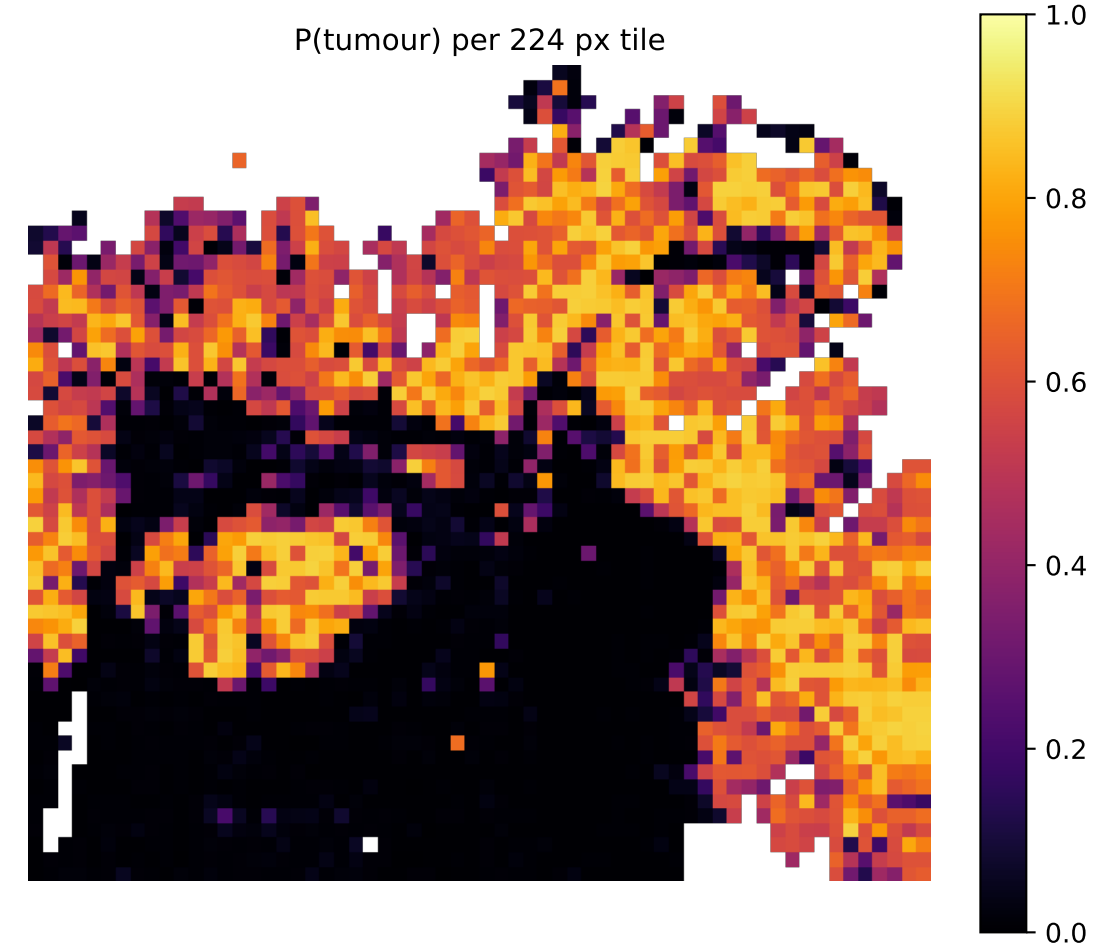
Cancer_P2 -- H&E (ground truth: tumour)



tumour boundary at $P(\text{TUM}) \geq 0.4$ -- 1 region(s)

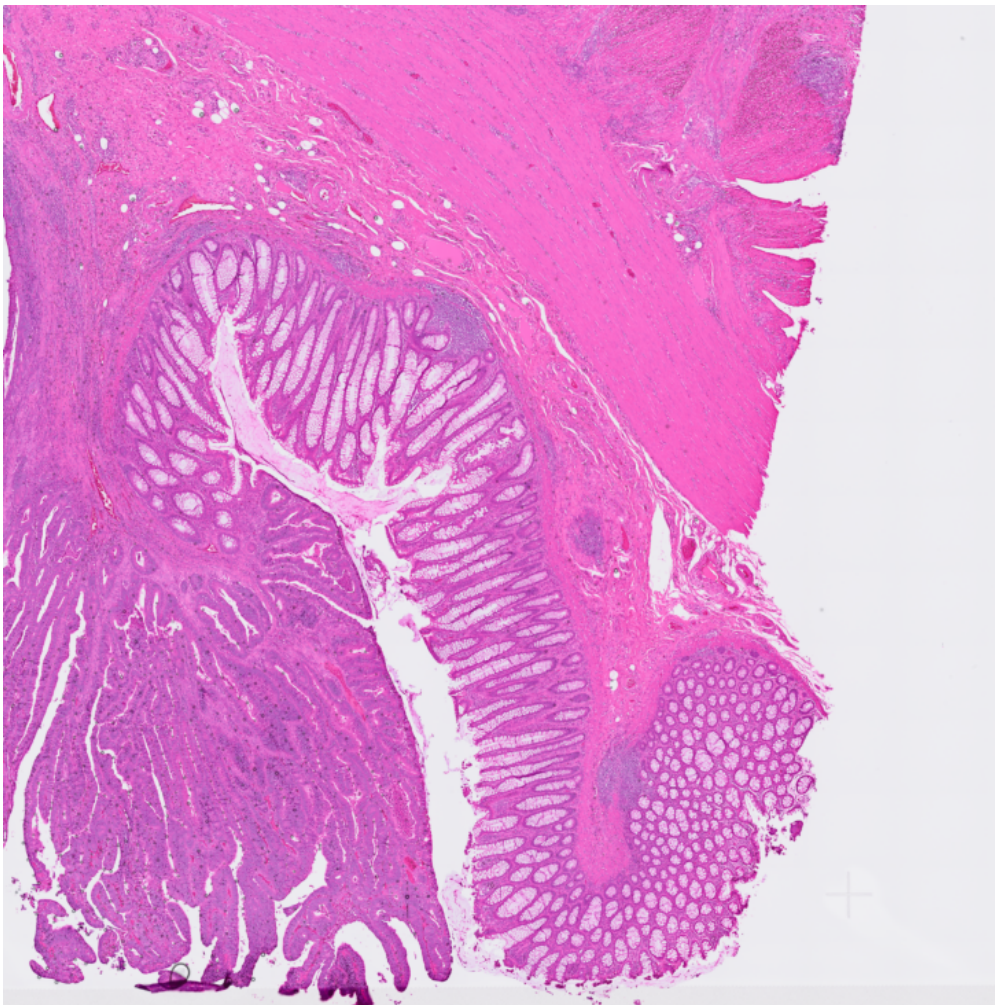


$P(\text{tumour})$ per 224 px tile

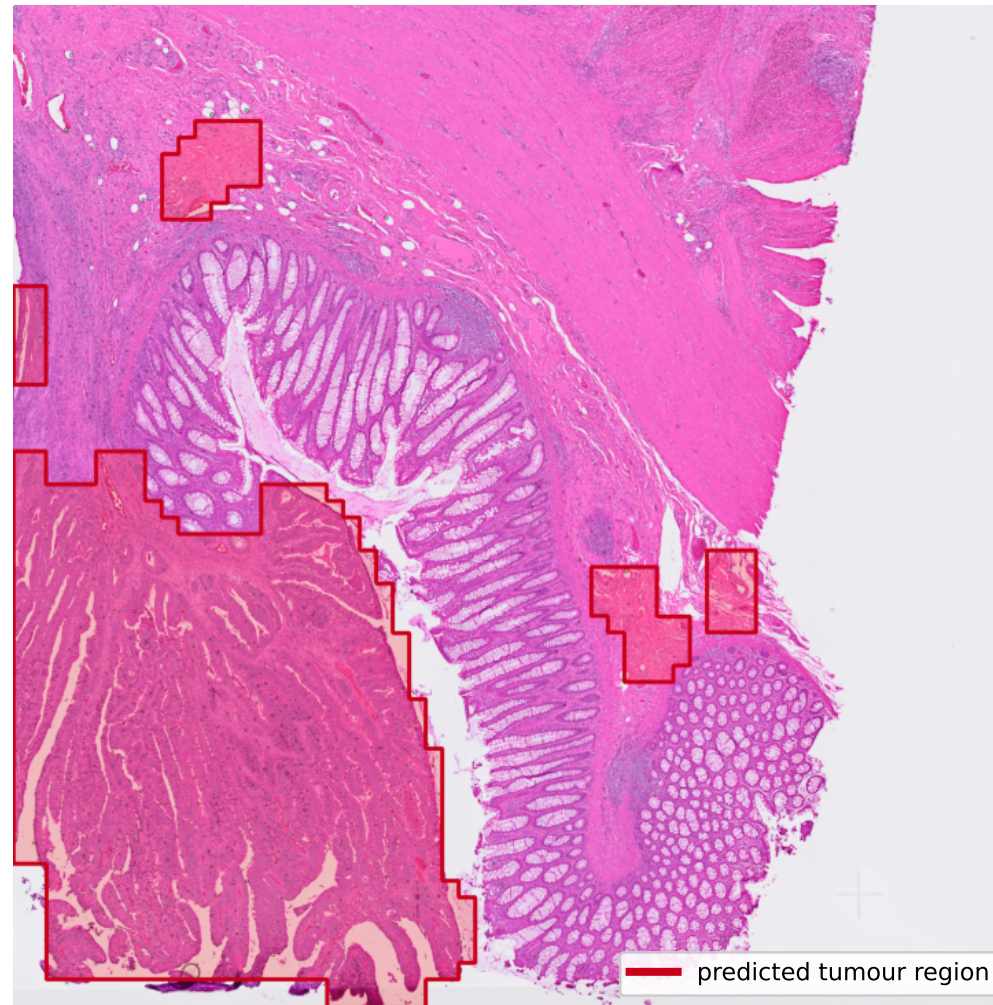


Cancer_P5 | tumour area 27.7% | 2,794 tissue tiles | exported regions 5 | correctly outlines tumour

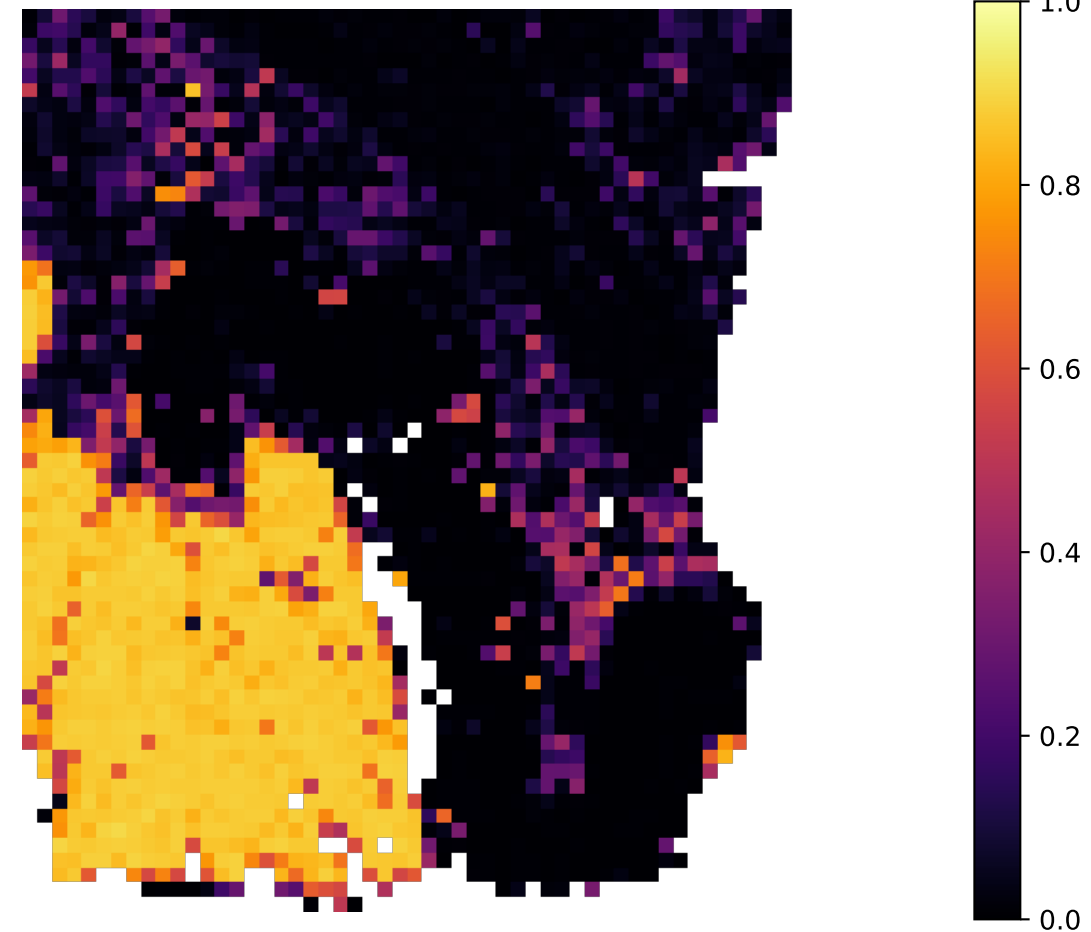
Cancer_P5 -- H&E (ground truth: tumour)



tumour boundary at $P(\text{TUM}) \geq 0.4$ -- 5 region(s)

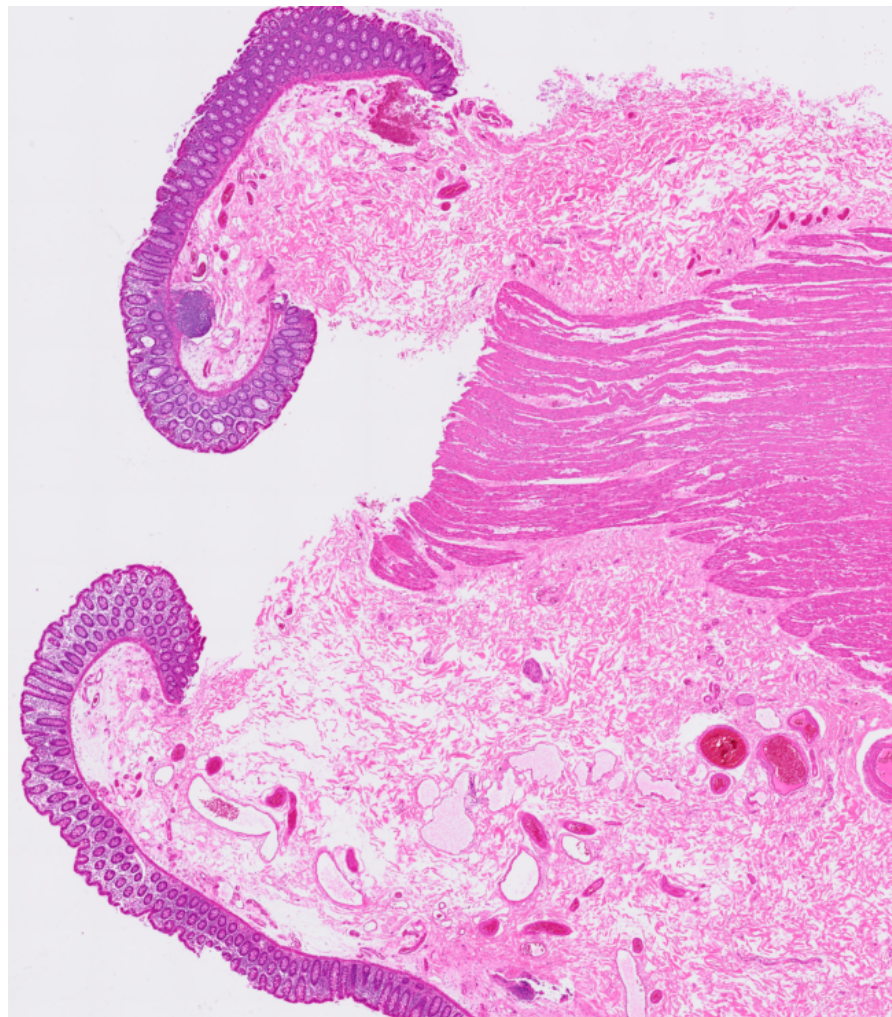


$P(\text{tumour})$ per 224 px tile

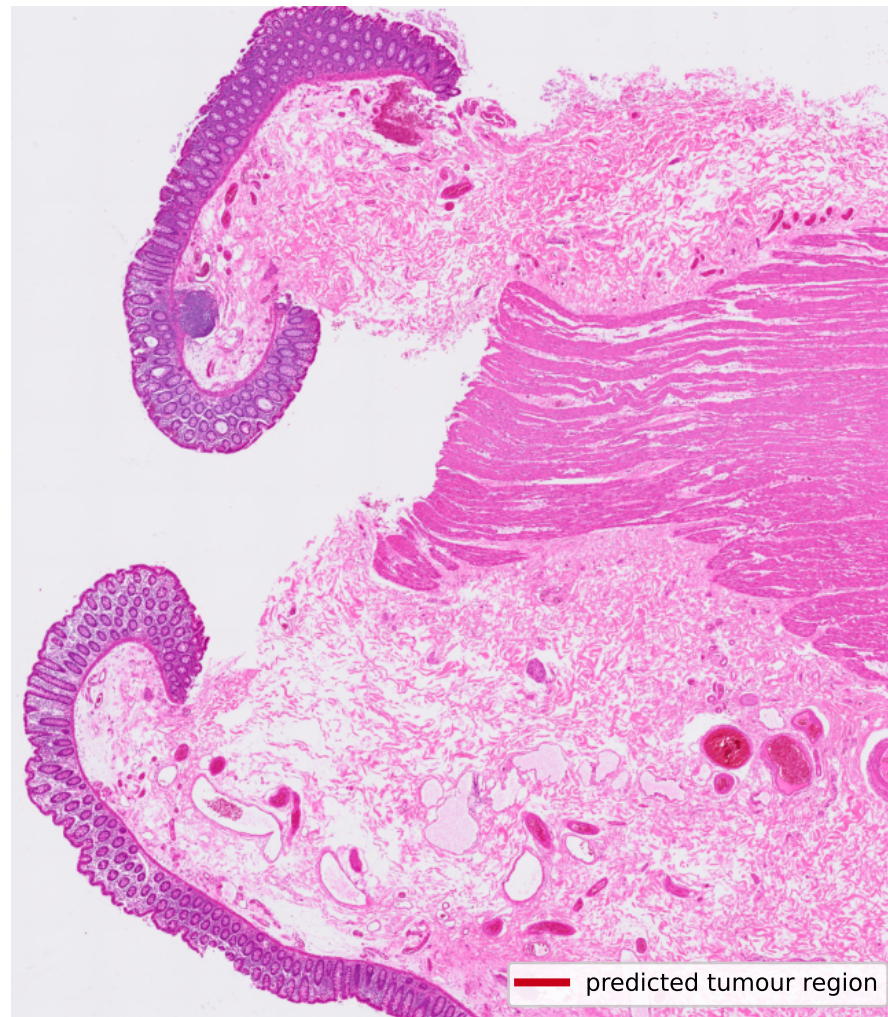


Normal_P3 | tumour area 0.1% | 2,454 tissue tiles | exported regions 0 | no tumour called -- correct

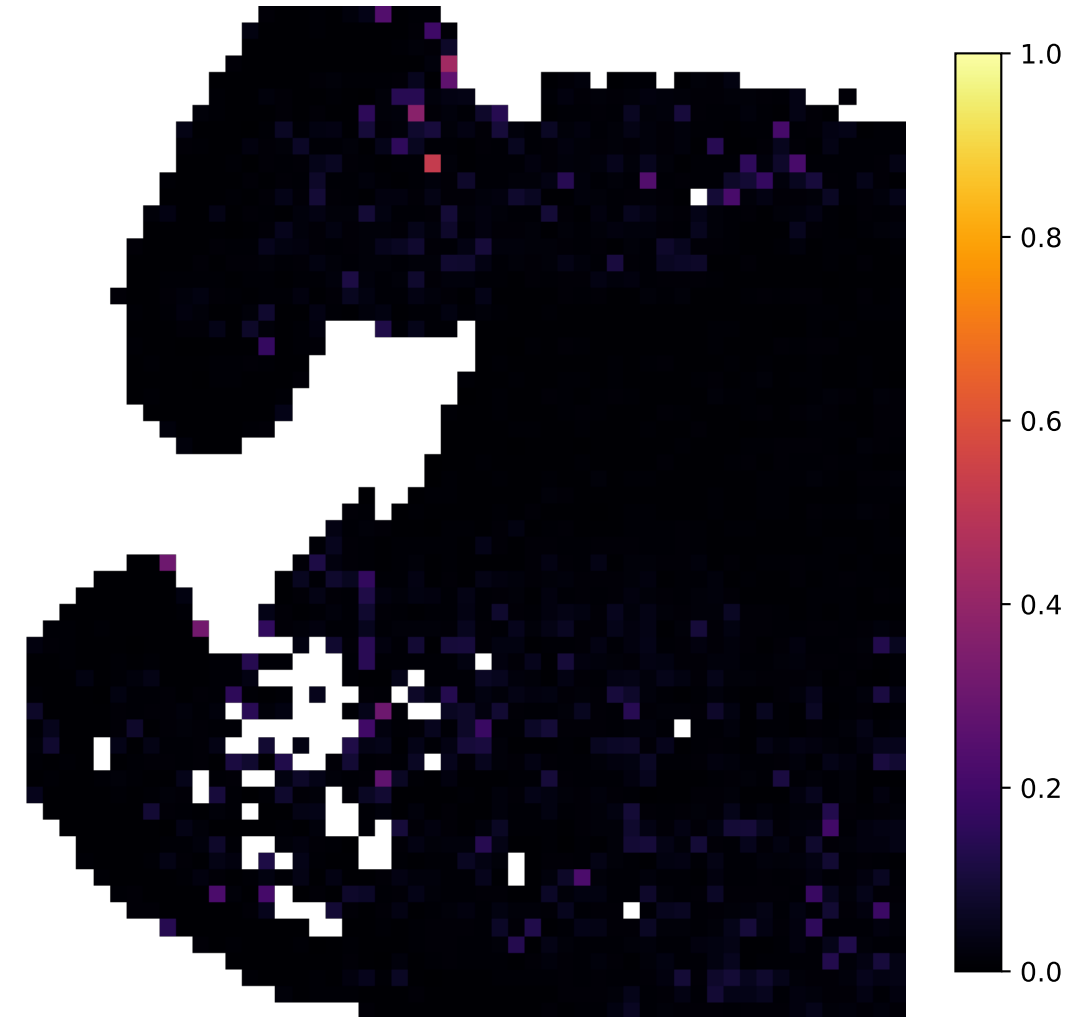
Normal_P3 -- H&E (ground truth: normal)



tumour boundary at $P(\text{TUM}) \geq 0.4$ -- 0 region(s)

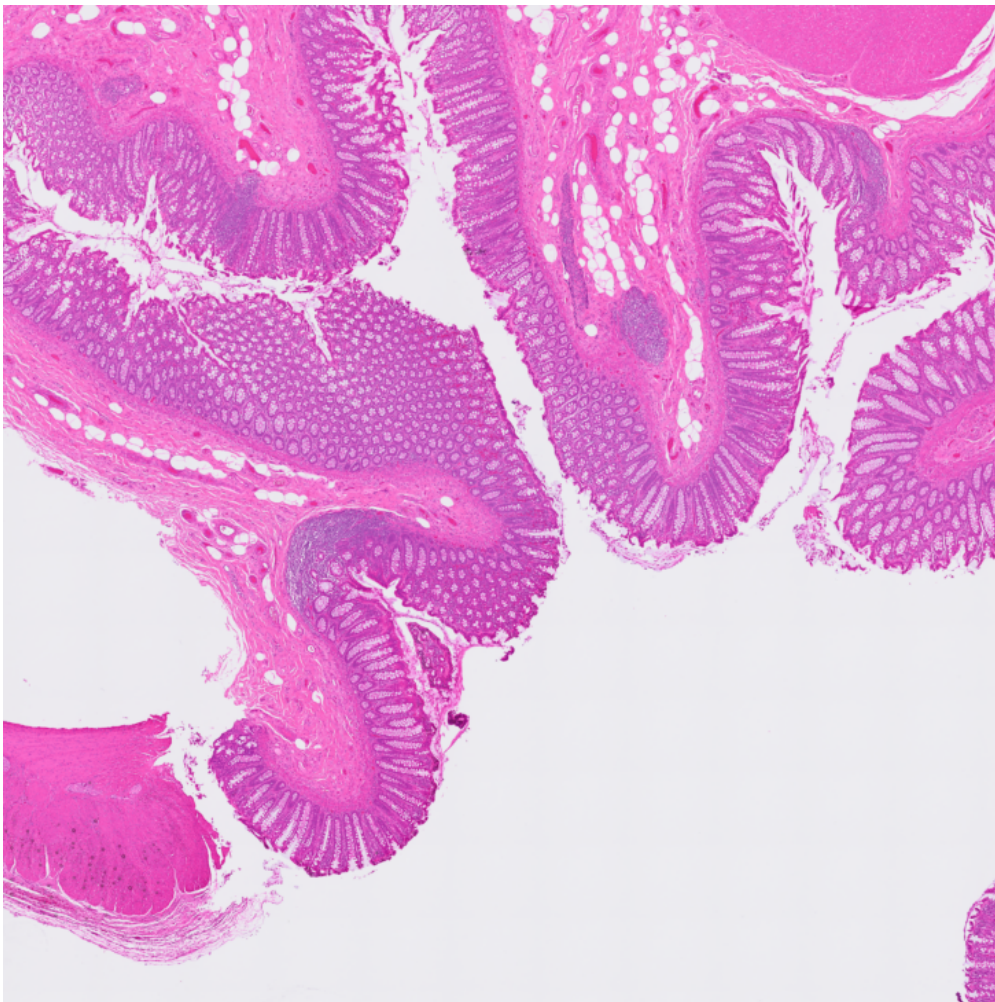


$P(\text{tumour})$ per 224 px tile

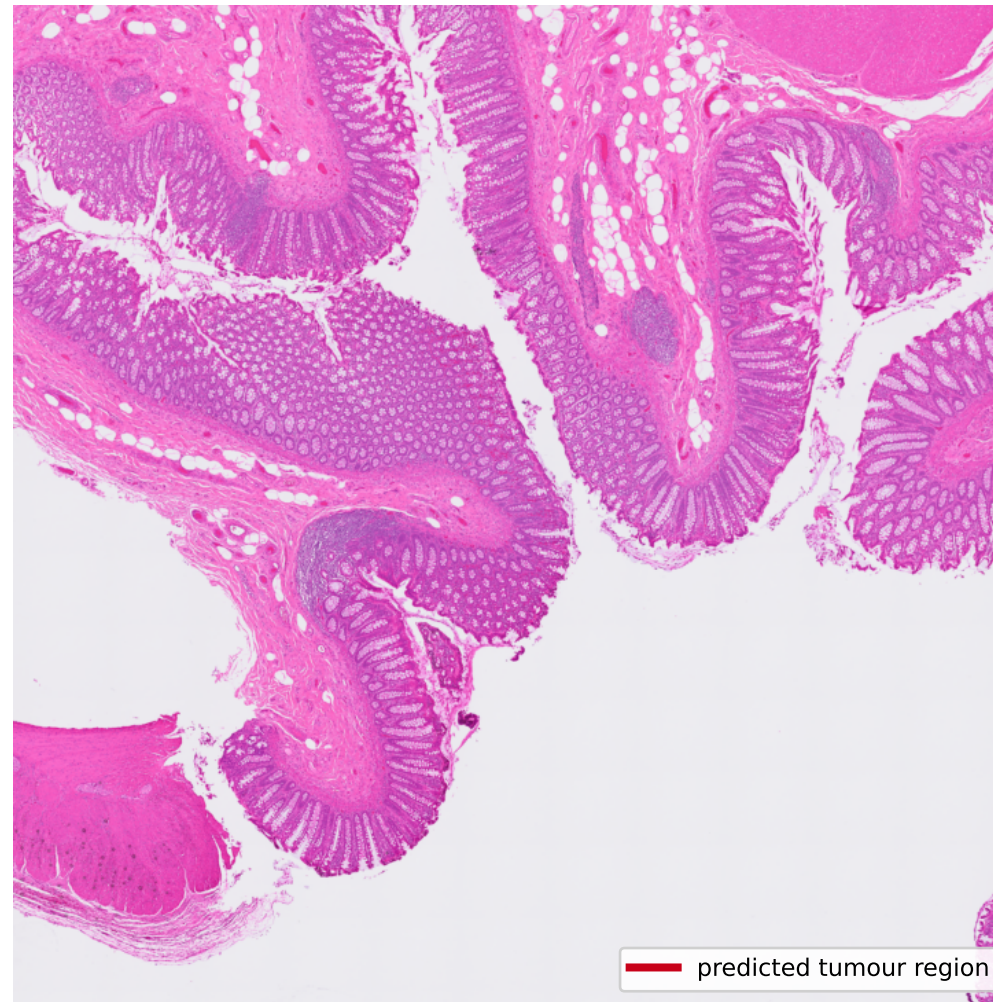


Normal_P5 | tumour area 1.8% | 2,365 tissue tiles | exported regions 0 | no tumour called -- correct

Normal_P5 -- H&E (ground truth: normal)



tumour boundary at $P(\text{TUM}) \geq 0.4$ -- 0 region(s)



$P(\text{tumour})$ per 224 px tile

